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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Ideal Gas

(English Medium)

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IDEAL GAS

1. INTRODUCTION

Matter, as we know, broadly exist in three states - solid, liquid and gas.

There are always two opposite tendencies between particles of matter which determine the state of matter :

- Intermolecular forces.
- The molecular motion / random motion (energy of particles)

Intermolecular forces are the forces of attraction and repulsion between atoms or molecules. Attractive intermolecular forces are known as ***vander Waals forces***. These are dispersion forces, dipole-dipole forces & dipole induced forces. When two molecules are brought very close, they will exert repulsive forces. Magnitude of the repulsion rises very rapidly as the distance separating the molecules decreases. This is the reason that liquid & solids are hard to compress. In these states, molecules are already in close contact, therefore they resist further compression (in that case repulsive interaction will increase) Thermal energy is the energy of a body arising from motion of its atoms or molecules. It is directly proportional to the temperature of the substance. It is the measure of average kinetic energy of the particles of the matter and is thus responsible for movement of particles. This movement of particles is called ***thermal motion***. Intermolecular forces tend to keep the molecules together but thermal energy of the molecules tends to keep them apart. Three states of matter are the result of balance between intermolecular forces and the thermal energy of the molecules.

2. GENERAL CHARACTERISTICS OF SOLID, LIQUID & GAS

Each physical state of matter possesses characteristics properties of its own. For example,

❖ **Solids :**

All solids show the following characteristics :

(i) Solids are rigid and incompressible.

(ii) Solids have fixed shape and definite volume.

(iii) Solids have their melting and boiling points above room temperature.

(iv) Density of solid is high.

❖ **Liquids :**

All liquids show the following characteristics :

(i) Liquids are almost incompressible but less incompressible than solids.

(ii) Liquids have fixed volume but no fixed shape.

(iii) Liquids have their melting points below room temperature and boiling points above room temperature, under normal conditions.

(iv) Density of liquids is lower than that of solids but much higher than that of gases.

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❖ **Gases :**

All gases show the following characteristics :

(i) Gases are highly compressible, i.e. gases can be compressed easily by applying pressure.

(ii) Gases have no fixed volume and shape. Gases fill the container of any size and shape completely.

(iii) Gases can diffuse into each other rapidly.

(iv) Gases have their melting and boiling points both below room temperature.

(v) Gases generally have low density.

3. MEASURABLE PROPERTIES OF GASES

The characteristics of gases are described fully in terms of four parameters or measurable properties :

(I) Amount of the gas (i.e., mass or number of moles).

(II) Volume (V) of the gas.

(III) Temperature (T)

(IV) Pressure (P)

I. Amount of the gas :

(i) The mass of a gas can be determined by weighing the container in which the gas is enclosed and again weighing the container after removing the gas. The difference between the two masses gives the mass of the gas.

Mass of gas (m) = Mass of filled container – mass of empty container

(ii) The mass of the gas is related to the number of moles of the gas as

$$\textit{Moles of gas (n)} = \frac{\text{Mass (in gm)}}{\text{Molar mass}} = \frac{m}{M}$$

(iii) Mass is expressed in gram or kg.

II. Gas volume :

(i) Since gases occupy the entire space available to them, the measurement of volume of a gas only requires a measurement of the capacity of container confining the gas.

(ii) Volume is expressed in litres (L), millilitres (mL), cubic centimeters (cm^3), cubic meters (m^3), etc.

(iii) $1\ L = 1000\ mL = 1\ dm^3 = 10^{-3}\ m^3$

$1\ m^3 = 10^3\ dm^3 = 10^6\ cm^3 = 10^6\ mL = 10^3\ L$

$1\ mL = 1\ cc = 1\ cm^3$

III. Temperature :

(ii) Temperature is measured in degree Celsius (°C), degree Fahrenheit (°F), Kelvin (K), etc.

(iii) S.I. unit of temperature is kelvin (K)

$K = °C + 273.15$

(iv) Relation between °F and °C is

$$\frac{°C}{5} = \frac{°F - 32}{9}$$

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IV. Pressure :

Pressure is the force exerted by the gas per unit area on the walls of the container in all directions. Thus,

Pressure (P) = Force(F) / Area(A)

❖ **Atmospheric pressure :**

The pressure exerted by atmosphere on earth's surface at sea level is called atmospheric pressure. Generally it is expressed in atm.

1 atm = 1.013 bar = 1.013×10^5 N/m^2 = 1.013×10^5 Pa = 1.013×10^6 dyne/cm^2

1 atm = 76 cm of Hg = 760 mm of Hg = 760 torr

1 bar = 75 cm Hg.

4. GAS LAWS

The behaviour of the gases is governed by same general laws, which were discovered as a result of their experimental studies. These laws are relationships between measurable properties of gases. Some of these properties like pressure, volume, temperature and mass are very important because relationships between these variables describe state of the gas.

The first reliable measurement on properties of gases was made by Anglo-Irish scientist Robert Boyle in 1662. The law which he formulated is known as ***Boyle's Law***. Later on attempts to fly in air with the help of hot air balloons motivated Jaccques Charles and Joseph Lewis Gay Lussac to discover additional gas laws. Contribution from *Avogadro* and others provided lot of information about gaseous state.

4.1 Boyle's Law :

For a fixed amount of gas at constant temperature, the volume occupied by the gas is inversely proportional to the pressure applied on the gas or pressure of the gas.

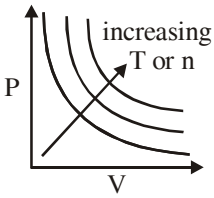
$\Rightarrow \quad V \propto \frac{1}{P}$

$\Rightarrow \quad PV = \text{constant}$

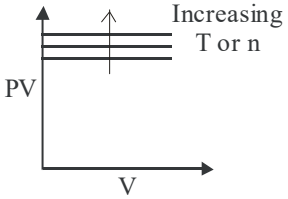
$\Rightarrow$ Working formula : $\boxed{P_1V_1 = P_2V_2}$

❖ **Graphical representation of Boyle's law (Isotherms) :**

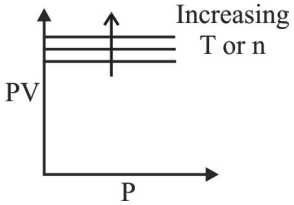
(i) P v/s V :



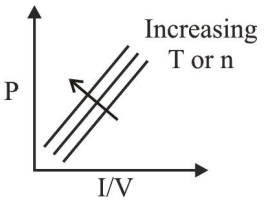
(ii) PV v/s V :



(iii) PV v/s P :



(iv) P v/s 1/V :



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Ex.1 ***A sample of gas occupies 100 dm^3 at 1 bar pressure and at T °C. If the volume of the gas is reduced to 5 dm^3 at the same temperature, what additional pressure must be applied ?***

Sol. From the given data :

$P_1 = 1$ bar $\quad P_2 = ?$

$V_1 = 100\ dm^3 \qquad V_2 = 5\ dm^3$

Since the temperature is constant, Boyle's law can be applied

$$P_1V_1 = P_2V_2 = \quad P_2 = \frac{P_1V_1}{V_2}$$

$$P_2 = \frac{(1\,bar) \times (100\,dm^3)}{(5\,dm^3)} = 20\,bar$$

$\therefore$ Additional pressure applied = 20 – 1 **= 19 bar**

4.2. Charle's Law :

For a fixed amount of gas at constant pressure, volume occupied by the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$\Rightarrow \quad V \alpha T$

$$\boxed{\frac{V}{T} = \text{constant}}$$

T = Temperature on absolute scale, kelvin scale or ideal gas scale.

Working formula : $\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$

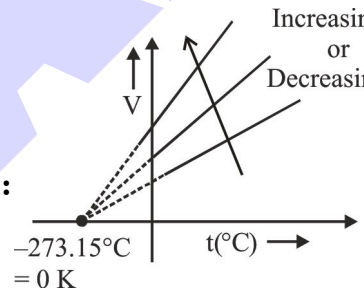
$* \qquad V = V_0\left(1 + \frac{t}{273.15}\right)$

t = temperature on celcius scale.

V_0 = volume of gas at 0°C.

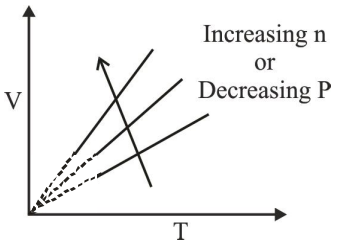
❖ **Graphical representation of Charle's Law (Isobars) :**

(i) V v/s t (°C) :



= Absolute zero

(ii) V v/s T :



❖ **Important Points :**

- Absolute zero (0 K = –273.15°C) is theoritically the minimum possible temperature.
- Absolute zero can never be achieved.

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Ex.2 ***A certain amount of a gas at 27°C and 1 bar pressure occupies a volume of 25 m^3. If the pressure is kept constant and the temperature is raised to 77°C, what will be the volume of the gas ?***

Sol. From the available data :

$V_1 = 25\ m^3$ $T_1 = 27 + 273 = 300\ K$

$V_2 = ?$ $T_2 = 77 + 273 = 350\ K$

Since the pressure of the gas is constant, Charles law is applicable

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$V_2 = \frac{V_1 \times T_2}{T_1} = \frac{(25 m^3) \times (350 K)}{(300 K)} = \mathbf{29.17\ m^3}$$

Ex.3 ***A balloon is inflated to $\frac{7}{8}$ of its maximum volume at 27°C then calculate the minimum temperature above which it will burst.***

Sol. $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{7V}{8 \times 300} = \frac{V}{T}$$

$T = 342.8\ K$

4.3. Gay-lussac's law or Pressure law :

For a fixed amount of gas at constant volume, pressure of the gas is directly proportional to temperature of the gas on absolute scale of temperature.

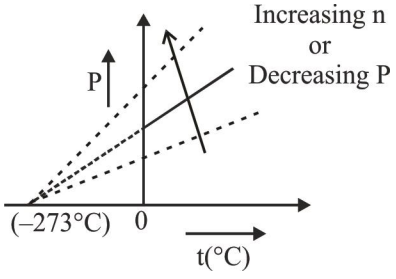
$\Rightarrow$ $P \propto T$

$$\frac{P}{T} = \text{constant}$$

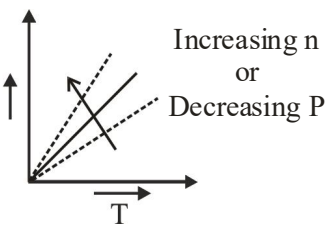
Working formula $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

* $$P = P_0\left(1 + \frac{t}{273.15}\right)$$

(i) P vs t (°C) :



(ii) P vs T :



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Ex.4 ***A gas cylinder containing cooking gas can with stand a pressure of 18 atm. The pressure gauge of cylinder indicates 12 atm at 27°C. Due to sudden fire in building the temperature start rising at what temperature will the cylinder explode.***

Sol. $\frac{P_1}{T_2} = \frac{P_2}{T_2}$

$\frac{12}{300} = \frac{18}{T}$

$T = 450\ K$

4.4. Avogadro's law :

Equal volumes of all the gases under similar conditions of temperature and pressure contains equal number of molecules or moles of molecules (not atoms).

$V \propto N$ (Temperature and pressure constant)

$V \propto n$ (Temperature and pressure constant)

Where, N = number of molecules, n = number of moles of molecules

$$\frac{V_1}{N_1} = \frac{V_2}{N_2} \text{ or } \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Since volume of a gas is directly proportional to the number of moles; one mole of each gas at standard temperature and pressure (STP) will have same volume.

Standard temperature and pressure means 273.15 K (0°C) temperature and 1 bar (i.e., exactly 10^5 pascal) pressure. **At STP, molar volume of an ideal gas or a combination of ideal gases is 22.71098 L mol^{-1}, normally taken as 22.7 L mol^{-1}.**

* The previous standard is still often used, and applies to all chemistry data more than decade old. In this definition STP denotes the same temperature of 0°C (273.15 K), but a slightly higher pressure of 1 atm (101.325 kPa). One mole of any gas or a combination of gases occupies 22.413996 L of volume at STP, normally taken as 22.4 L.

Standard ambient temperature and pressure (SATP), conditions are also used in some scientific works. SATP conditions means 298.15 K and 1 bar (i.e., exactly 10^5 Pa). **At SATP (1 bar and 298.15 K), the molar volume of an ideal gas is 24.789 L mol^{-1}.**

Molar volume in litres per mole of some gases at 273.15 K and 1 bar (STP).	
Argon	22.37
Carbon dioxide	22.54
Dinitrogen	22.69
Dioxygen	22.69
Dihydrogen	22.72
Ideal gas	22.71

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Ex.5 A toy balloon originally held 1.0 gm of He gas and had a radius 10 cm. During the night, 0.875 gm of the gas effused from the balloon. Assuming ideal gas behaviour, under these constant P and T conditions, what was the radius of the balloon the next morning.

Ans. 5 cm

Sol. $\frac{V_1}{n_1}=\frac{V_2}{n_2} \Rightarrow \frac{V_1}{m_1}=\frac{V_2}{m_2}$

$$\Rightarrow \frac{\frac{4}{3}\pi(10)^3}{1.0}=\frac{\frac{4}{3}\pi r^3}{1.0-0.875}$$

$\therefore$ r = 5 cm.

5. IDEAL GAS EQUATION

A single equation which is combination of Boyle's, Charle's & Avogadro's Law is known as ***Ideal gas equation***, or we can say a single equation which describe the simultaneous effects of the change in temperature and pressure on volume of the given amount of the gas is called the ***Ideal gas equation.***

According to Boyle's law, $V \propto \frac{1}{P}$ (at constant T and n)

According to Charle's law, $V \propto T$ (at constant P and n)

According to Avogadro's law, $V \propto n$ (at constant T and P)

Combining these three laws, $V \propto \frac{nT}{P}$ or $PV \propto nT$

or **PV = nRT** [Equation of state for ideal gas or combined gas equation or ideal gas equation]

where R is a constant called ***Universal gas constant*****,** which does not depend on variables (P, V, n, T) and nature of gas.

5.1 Molar volume is the volume of 1 mole of gas.

Molar volume $(V_m)=\frac{\text{Volume}}{\text{mole}}$

$$PV_m = RT$$

Volume of 1 mole of an ideal gas under STP conditions (273.15 K, normally taken as 273 K, and 1 bar pressure) is 22.7 L mol^{-1}.

5.2 Dimension of R :

$$R=\frac{PV}{nT}=\frac{\text{Pressure}\times\text{Volume}}{\text{Mole}\times\text{Temperature}}=\frac{(\text{Force}/\text{Area})\times(\text{Area}\times\text{Length})}{\text{Mole}\times\text{Temperature(K)}}$$

$$=\frac{\text{Force}\times\text{Length}}{\text{Mole}\times\text{Temperature(K)}}=\frac{\text{Work or energy}}{\text{Mole}\times\text{Temperature(K)}}$$

5.3 Physical significance of R :

The dimensions of R is energy per mole per kelvin and hence it represents the amount of work (or energy) that can be obtained from one mole of a gas when its temperature is raised by 1 K, isobarically.

5.4 Value of R is different units :

(i) In L-atm : $R = \frac{1\,atm \times 22.4\,lit.}{1\,mol \times 273\,K}$ **= 0.082 L-atm mol^{-1}K^{-1}**

(ii) In L-bar : $R = \frac{1\,bar \times 22.7\,L}{1\,mol \times 273\,K}$ **= 0.083 L-bar/K-mol**

(iii) In C.G.S system : $R = \frac{(1 \times 76 \times 13.6 \times 980\ dyne\,cm^{-2}) \times (22400\,cm^3)}{1\,mol \times 273\,K}$

= 8.314 × 10^7 erg mole^{-1} K^{-1}.

(iv) In SI units : **R = 8.314 Joule mole^{-1} K^{-1}.** [10^7 erg = 1 joule]

(v) In calories : R = 1.987 ≈ **2 calorie mole^{-1} K^{-1}.**

5.5 Ideal gas equation in terms of density of gas :

$$PV = nRT = \frac{W}{M}RT$$

or $$PM = \frac{W}{V}.RT$$

$\therefore$ **PM = dRT**

Ex.6 ***The best vacuum so far attained in laboratory is 10^{-10} mm of Hg. What is the number of molecules of gas remain per cm^3 at 20°C in this vacuum ?***

Sol. Given conditions -

$P = 10^{-10}\ mmHg = \frac{10^{-10}}{760}atm$; $V = 1\ cm^3 = \frac{1}{1000}L$; $T = 20°C = 293\ K$

No. of molecules, N = ?

Now, applying

$PV = nRT$

$\therefore$ number of molecules per cm^3 , $N = \frac{N_A \times P \times V}{RT}$

$\Rightarrow$ $N = \frac{6.022 \times 10^{23} \times (10^{-10}/760) \times (1/1000)}{(0.082)(293)}$

N = 3.29 × 10^6 molecules

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Ex.7 ***The density of a gas is found to be 1.56 g dm^{-3} at 0.98 bar pressure and 65°C. Calculate the molar mass of the gas.***
Use R = 0.083 dm^3 bar K^{-1} mol^{-1}

Sol. We know that :

$$M = \frac{dRT}{P} = \frac{(1.56\,g\,dm^{-3}) \times (0.083\,dm^3\,bar\,K^{-1}\,mol^{-1}) \times 338K}{(0.98\,bar)} = \mathbf{44.66\ g\ mol^{-1}}$$

Ex.8 ***A tyre tube of maximum volume 8.2 lit. can withstand a pressure of 10 atm. Initially the tube is empty.***
(i) ***Calculate the number of moles required to inflate completely the tube upto a pressure of 1 atm & 300 K temperature.***
(ii) ***Calculate the minimum number of moles required to burst the tyre tube at 300 K.***

Sol. **(i)** $PV = nRT$

$1 \times 8.2 = n_1 \times 0.082 \times 300$

$n_1 = \frac{1}{3}$

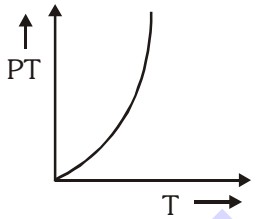
(ii) $PV = nRT$

$10 \times 8.2 = n_2 \times 0.082 \times 300$

$n_2 = \frac{10}{3}$

Ex.9 ***(a) Plot the curve between PT vs T at const V & constant no. of moles.***
(b) Find the number of moles of gas taken when the volume of the vessel is 82.0 ml and $\frac{d}{dT}$[PT] at 300 K = 300 for the given curve.

Ans. **(a)**

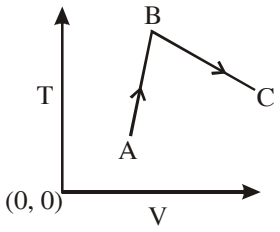


(b) $n = \frac{1}{2}$

Sol. (b) $\frac{d(PT)}{dT} = \frac{d}{dT}\left[\frac{nRT}{V}.T\right] = \frac{2nRT}{V} = 300$

$\therefore n = \frac{300 \times (82 \times 10^{-3})}{2 \times 0.082 \times 300} = \frac{1}{2}$

Ex.10 In the given isobaric process shown by graph between T & V.

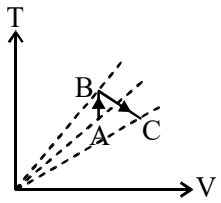


(A) Moles decreases throughout
(B) Moles first increases then decreases
(C) Moles first decreases then increases
(D) Moles cannot be predicted form given data

Ans. **(C)**

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Sol.



$$\frac{dT}{dV} = \frac{P}{nR}$$

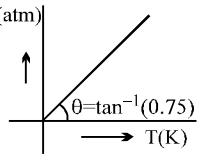
At constant pressures slope is higher for smaller moles of gas.

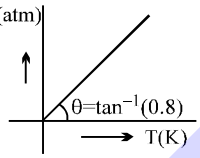
Hence, from A to B, mole of gas decreases and for B to C, moles increases.

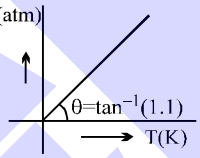
Ex.11 Gas A (1 mol) dissociates in a closed rigid container of volume 0.16 lit. as per following reaction.

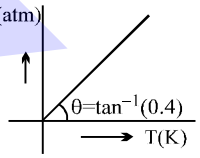
2A (g) ⟶ 3B (g) + 2C (g)

If degree of dissociation of A is 0.4 and remains constant in entire range of temperature, then the correct P vs T graph is [Given R = 0.08 lit-atm/mol/K]

(A) 

(B) 

(C) 

(D) 

Ans. (B)

Sol.

	2A	⟶	3B	+	2C
Initial	1		0		0
Final	1 – 0.4 = 0.6		0.6		0.4

Total moles = 0.6 + 0.6 + 0.4 = 1.6

Now, $\frac{dP}{dT} = \frac{nR}{V} = \frac{1.6 \times 0.08}{0.16} = 0.8 \tan\theta$

$\therefore \quad \theta = \tan^{-1}(0.8)$

Ex.12 LPG is a mixture of n-butane & iso-butane. What is the volume of oxygen needed to burn 1 kg of LPG at 1 atm, 273 K ?

Sol. During the burning of LPG following reaction takes place -

$$C_4H_{10} + \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5\,H_2O$$

Now, for complete combustion of 1 mole of LPG $\longrightarrow \frac{13}{2}$ moles of O_2 are required.

$\therefore$ for $\frac{1000}{58}$ moles of LPG $\longrightarrow \frac{13}{2} \times \frac{1000}{58}$ moles of O_2 are required.

Thus, the volume of oxygen needed to burn 1 kg of LPG at 1 atm & 273K would be

Vol. of O_2 = Moles of $O_2 \times 22.4$ L = $\frac{13}{2} \times \frac{1000}{58} \times 22.4$ = **2510 L**

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Ex.13 ***An open flask contains air at 27°C. Calculate the temperature at which it should be heated so that***

(a) ***1/3rd of air measured in the container at 27°C escape out.***

(b) ***1/3rd of air measured in the container at final temperature escape out.***

Sol. **(a)** $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times T}=\frac{1}{3}\times\frac{PV}{R\times 300} \Rightarrow \frac{1}{300}-\frac{1}{T}=\frac{1}{900}$$

$$\frac{1}{T}=\frac{1}{300}-\frac{1}{900}=\frac{3-1}{900}=\frac{2}{900} \Rightarrow T = 450\ K = 177°C$$

(b) $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times T}=\frac{1}{3}\times\frac{PV}{R\times T}$$

$$\frac{1}{300}-\frac{1}{T}=\frac{1}{3\times T}$$

$\Rightarrow$ T = 400 K = 127°C

Ex.14 ***A bulb of unknown volume containing air is heated from 27°C to 227°C at constant pressure. The expelled air is measured at different temperature of determine volume of container. What will be volume of container if -***

(a) ***200 ml of air measured at 227°C was expelled.***

(b) ***200 ml of air measured at 27°C was expelled.***

(c) ***200 ml of air measured at 127°C was expelled.***

Sol. **(a)** $\mathbf{n_{initial} - n_{final} = n_{expelled}}$

$$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 500}$$

$$\frac{V}{3}-\frac{V}{5}=\frac{200}{5}$$

$$\frac{5V-3V}{15}=\frac{200}{5}$$

$$\Rightarrow 2V=\frac{200\times 15}{5} \Rightarrow V = 300\ ml$$

(b) $$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 300} \Rightarrow \frac{2V}{15}=\frac{200}{3} \Rightarrow V=\frac{100\times 15}{2\times 3}=500ml$$

(c) $$\frac{PV}{R\times 300}-\frac{PV}{R\times 500}=\frac{P\times 200}{R\times 400}$$

$$\frac{2V}{15}=\frac{200}{4} \Rightarrow V=\frac{200\times 15}{2\times 4}=375ml$$

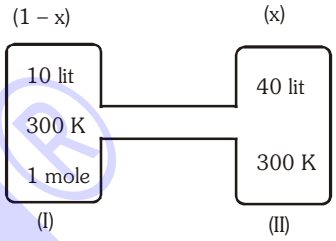
Ex.15 A 10 litre container consist of 1 mole of gas at 300 K. It is connected to another container having volume 40 litre and is initially at 300 K. The nozzle connecting two containers is opened for a long time and once the movement of gas stopped, the larger container was heated to a temperature of 600 K. Calculate

(a) Moles and pressure of gas in both the containers before heating.

(b) Moles and pressure in two containers after heating.

(Assume that initially the larger container is completely evacuated.)

Sol. **(a)** Before heating :



$P_I = P_{II}$

$$\frac{(1-x)R\times 300}{10} = \frac{x\times R\times 300}{40}$$

$x = 0.8$ moles

$n_I = 1 - x = 0.2$ mole

$n_{II} = x = 0.8$ mole

$$\text{Pressure} = \frac{x\times R\times T}{V} = \frac{0.8\times R\times 300}{40} = 0.492 \text{ atm}$$

(b) After heating :

$$\frac{(1-x_1)R\times 300}{10} = \frac{x_1\times R\times 600}{40}$$

$x_1 = 0.67$ moles, Given $T_1 = 600$ K

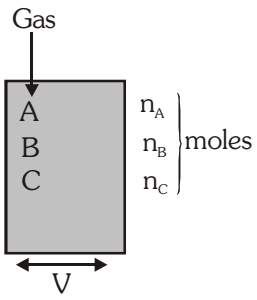
$$\text{Pressure} = \frac{x_1\times R\times T_1}{V} = \frac{0.67\times 0.082\times 600}{40} = 0.82 \text{ atm}$$

6. DALTON'S LAW OF PARTIAL PRESSURES

6.1 Partial pressure :

In a mixture of non-reacting gases, partial pressure of any component gas is defined as the pressure exerted by the individual gas if the entire volume of mixture had been occupied by this component only.

Partial pressure of component gases are -



$$P_A = \frac{n_A RT}{V} = \text{partial pressure of A}$$

$$P_B = \frac{n_B RT}{V} = \text{partial pressure of B}$$

$$P_C = \frac{n_C RT}{V} = \text{partial pressure of C}$$

6.2 Dalton's Law :

Dalton's law of partial pressure states "at a given temperature, the total pressure exerted by two or more non-reacting gases occupying a definite volume is equal to the sum of the partial pressures of the component gases."

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$P_{Total} = p_1 + p_2 + p_3 + ...$ (At constant V and T)

$$= \left(\frac{n_1}{V} + \frac{n_2}{V} + \frac{n_3}{V} + ...\right) RT = (n_1 + n_2 + n_3 + ...)\ \frac{RT}{V} = \frac{nRT}{V}$$

where $n = n_1 + n_2 + n_3 + ... =$ Total moles, V = Total volume

$$P_{Total} = \sum p_i = \frac{RT}{V}\sum n_i$$

- **Note : Dalton's law of partial pressure is applicable only to non-reacting gaseous mixture.**

6.3 Relation between partial pressure and total pressure :

If the two non-reacting gases A and B having n_A and n_B number of moles respectively are filled in a vessel of volume V at temperature T, then

$$\frac{p_A}{P} = \frac{n_A RT/V}{(n_A + n_B)RT/V} = \frac{n_A}{n_A + n_B} = x_A \text{ (mole fraction of A)}$$

$p_A = x_A \times P$, Similarly $p_B = x_B \times P$

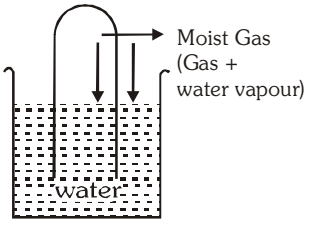
Partial pressure of a component = Mole fraction × total pressure.

6.4 It has been observed that gases are generally collected over water and therefore are moist.

$$\mathbf{P_{dry\ gas} = P_{moist\ gas} - P_{water\ vapour}}$$

or Pressure of dry gas = Pressure of moist gas – aqueous tension

The pressure exerted by water vapour is constant when it is in equilibrium with liquid water at a particular temperature. It is called vapour pressure of water or ***aqueous tension***, which varies with the temperature and becomes 760 mm at 100°C.



6.5 Relative Humidity (RH) $= \dfrac{\text{Partial pressure of water in air}}{\text{Vapour pressure of water (aq.tension)}}$

Ex.16 Calculate the total pressure in a 10 litre cylinder which contains 0.4 g He, 1.6 g oxygen and 1.4 g of nitrogen at 27°C. Also calculate the partial pressure of He gas in the cylinder. Assume ideal behavious for gases. ***[JEE 1997]***

Ans. 0.492 atm ; 0.246 atm

Sol. Given, V = 10 litre, T = 27 + 273 = 300K

Mole of He $= \frac{0.4}{4} = 0.1$; mole of $O_2 = \frac{1.6}{32} = 0.05$;

Mole of $N_2 = \frac{1.4}{28} = 0.05$

$\therefore$ Total moles = 0.1 + 0.05 + 0.05 = 0.20

$$P_M = \frac{nRT}{V} = \frac{0.2 \times 0.082 \times 300}{100} = 0.492 \text{atm}$$

$$\therefore\ P'_{He} = P_M \times \text{Mole fraction of He} = 0.492 \times \frac{0.05}{0.2} = 0.246 \text{ atm}$$

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Ex.17 (a) Find the total pressure and partial pressure of each component if a container of volume 8.2 lit. contains 2 moles of A and 3 mole of B at 300K.

(b) What will be the final pressure and partial pressure of each component if 5 moles of C is also added to the container at same temperature.

Sol. (a) $P_A = \frac{n_A RT}{V} = \frac{2 \times 0.082 \times 300}{8.22} = 6\,atm$

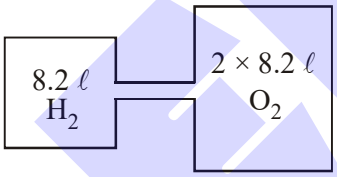
$P_B = \frac{n_B RT}{V} = \frac{3 \times 0.082 \times 300}{8.22} = 9\,atm$

Total pressure $P_T = P_A + P_B = 6 + 9 = 15$ atm

(b) $P_C = \frac{n_C RT}{V} = \frac{5 \times 0.082 \times 300}{8.22} = 15\,atm$

and total pressure = 15 + 15 = 30 atm.

Ex.18 A container of 8.2 lit. capacity is filled with 1 mole of H_2 at 300 K and it is connected to another container of capacity 2 × 8.2 lit. containing 4 moles of O_2 at 300 K, then find the final pressure & partial pressure of each gas.



Sol. $P_f V_f = n_f RT$

$P_f (3 \times 8.2) = 5 \times 0.082 \times 300$

$P_f = 5$ atm

$P_{H_2} = x_{H_2} P_f = \frac{1}{1+4} \times 5 = 1\,atm$

$P_{O_2} = x_{O_2} P_f = \frac{4}{1+4} \times 5 = 4\,atm$

Ex.19 2 moles of $NH_3(g)$ and 1 mole of HCl(g) are taken in a container of capacity 8.2 lit at 300K to produce $NH_4Cl(s)$. Find the total pressure after the reaction.

Sol.

	$NH_3(g)$	+ $HCl(g)$	$\longrightarrow$ $NH_4Cl(s)$
initial	2	1	
after rxn	1	0	

In this reaction HCl is L.R. so it will be completely consumed. We don't consider pressure due to solid.

$\therefore \quad P_T = \frac{nRT}{V} = \frac{1 \times 0.082 \times 300}{8.2} = 3\,atm$

Ex.20 A closed container containing O_2 and some liquid water was found to exert 740 mm pressure at 27°C.

(a) Then calculate the pressure exerted by O_2 if aqueous tension at 27°C is 20 mm.

(b) What will be the final pressure if volume is reduced to half.
(consider volume of liquid water negligible)

(c) What will be the final pressure if volume is doubled.

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Sol. (a) $P_T = P_{dry\ gas} + P_{aq.\ tension}$

$740 = P_{O_2} + 20$

$P_{O_2} = 740 - 20 = 720$ mm

(b) $(P_{O_2}V_{O_2})_{initial} = (P_{O_2}V_{O_2})_{final}$ (Boyle's law)

$720 \times V = P_{O_2} \times \frac{V}{2}$

$P_{O_2} = 1440\,mm$

$P_T = P_T = P_{O_2} + P_{aq.}$

$= 1440 + 20 = 1460$ mm

(c) $(P_{O_2}V_{O_2})_{initial} = (P_{O_2}V_{O_2})_{final}$

$720 \times V = P_{O_2} \times 2V$

$P_{O_2} = 360$ mm

$P_T = P_{O_2} + P_{aq.}$

$= 360 + 20 =$ **380 mm**

7. AMAGAT'S LAW OF PARTIAL VOLUME

7.1 Partial Volume :

Partial volume of any component is defined as the volume occupied by that particular component when it is kept at same total pressure and temperature as of the mixture.

7.2 Amagat's law :

According to this law at constant temperature and pressure, the total volume of mixture of non-reacting gases is equal to the sum of partial volumes of each component present in mixture.

$$V_T = V_1 + V_2 + V_3 + \ldots\ldots$$

Let us consider three non-reacting gases A, B and C are present in a container which have no. of moles n_A, n_B and n_C respectively. For each gas partial volume is

$$V_A = n_A\left(\frac{RT}{P}\right) = \text{partial volume of A}$$

$$V_B = n_B\left(\frac{RT}{P}\right) = \text{partial volume of B}$$

$$V_C = n_C\left(\frac{RT}{P}\right) = \text{partial volume of C}$$

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Now, total volume

$$V_T = V_A + V_B + V_C = (n_A + n_B + n_C)\left(\frac{RT}{P}\right) = n_T\left(\frac{RT}{P}\right)$$

$$\frac{V_A}{V_T} = \frac{n_A}{n_T} = x_A \qquad \text{(Mole fraction of gas A)}$$

$$\frac{V_B}{V_T} = \frac{n_B}{n_T} = x_B \qquad \text{(Mole fraction of gas B)}$$

$$\frac{V_C}{V_T} = \frac{n_C}{n_T} = x_C \qquad \text{(Mole fraction of gas C)}$$

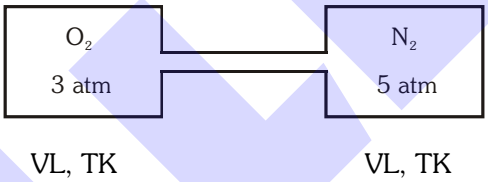
$\therefore$ ***Partial volume of a gas = Mole fraction × Total volume***

8. GRAHAM'S LAW OF DIFFUSION AND EFFUSION

8.1 Diffusion :

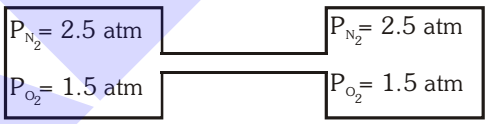
Diffusion is the process of gradual mixing of molecules of one gas with molecules of another gas due to their molecular motion (kinetic energy). The diffusion always proceeds from a region of high concentration to a region of lower concentration (or high partial pressure to low partial pressure). For example, when a bottle of perfume is opened at one end of the room, the person sitting at the other end of the room can smell the perfume because of the diffusion process of perfume molecules.

Note : Initially



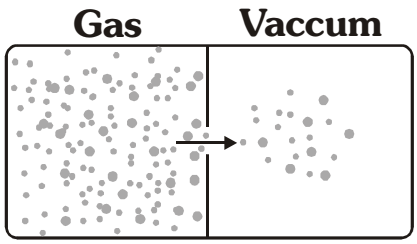
When stop cock is removed flow will be from both sides, N_2 will try to equalise its partial pressure in both the vessels, and so will O_2.

Finally



8.2 Effusion :

The effusion is the process of forcing a gas through a pin hole or small orifice from one compartment to another empty (vacuum) compartment.



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8.3 Graham's Law :

Under similar condition of pressure (partial pressure) and temperature, the rate of diffusion or effusion of different gases is inversely proportional to square root of their density.

$\Rightarrow$ rate of diffusion, $r \propto \frac{1}{\sqrt{d}}$

$\Rightarrow$ $\frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}} = \frac{\sqrt{V.D_2}}{\sqrt{V.D_1}}$

where, d = density of gas
V.D = vapour density
M = molar mass of gas

8.4 Under conditions of same temperature but different pressure :

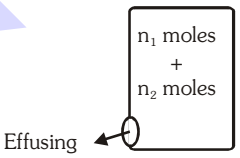
$r \propto \frac{P}{\sqrt{M}}$

$$\boxed{\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}}$$

If both gases are present in the same container at same temperature.

$P \propto n$

$\Rightarrow$ $\frac{r_1}{r_2} = \frac{n_1}{n_2}\sqrt{\frac{M_2}{M_1}}$



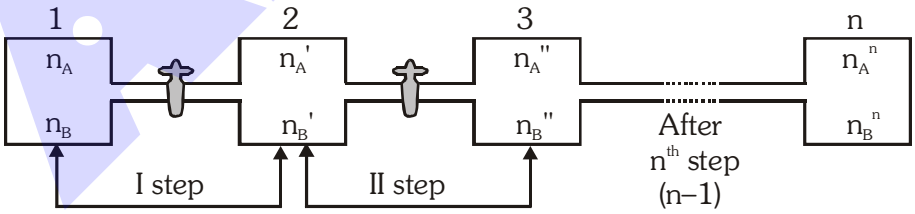
8.5 Rate of diffusion/effusion can be expressed as -

$r = \frac{\text{volume diffused}}{\text{time taken}}$ or $\frac{\text{moles diffused}}{\text{time taken}}$ or $\frac{\text{pressure dropped}}{\text{time taken}}$

or $\frac{\text{distance travelled in horizontal tube of uniform cross-section}}{\text{time taken}}$

8.6 Enrichment factor or Isotopic separation factor :

When two gases present in a container are allowed to diffuse in another container then gas having lower molecular mass will diffuse more and if this process is continued for large number of steps then we can obtain a mixture which is rich in a gas having lower molecular mass. Hence ultimately in the ultimate (last) container amount of lighter gas is larger as compared to heavier gas.



where, $M_A > M_B$ [similar condition of T and V] & $n_B^n >>>> n_A^n$

Enrichment factor (f) is defined as the ratio of final ratio of moles in the mixture [after n^{th} step] with initial mole ratio of the mixture.

$$f = \frac{\left(\frac{n_A^n}{n_B^n}\right)}{\left(\frac{n_A}{n_B}\right)} = \left(\sqrt{\frac{M_B}{M_A}}\right)^n = \frac{\text{final molar ratio}}{\text{initial molar ratio}}$$

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Ex.21 ***32 ml of He effuses through a fine orifice in 1 minute. Then what volume of CH_4 will diffuse in 1 minute under the similar condition.***

Sol. $\because \quad r \propto \frac{1}{\sqrt{M}}$

$$r = \frac{\text{volume diffused}}{\text{time}}$$

$\because$ time is same so

$$\frac{V_{CH_4}}{V_{He}} = \sqrt{\frac{M_{He}}{M_{CH_4}}}$$

$$\frac{V_{CH_4}}{32} = \sqrt{\frac{4}{16}}$$

$$V_{CH_4} = \frac{1}{2} \times 32 = 16\,mL$$

Ex.22 ***20 dm^3 of Ne diffuse through a porous partition in 60 seconds. What volume of SO_3 will diffuse under similar conditions in 30 sec. (Atomic wt. of Ne = 20, S = 32)***

Sol. $$\frac{r_{N_e}}{r_{SO_3}} = \sqrt{\frac{M_{SO_3}}{M_{Ne}}} \Rightarrow \frac{V_{Ne}/t_{Ne}}{V_{SO_3}/t_{SO_3}} = \sqrt{\frac{M_{SO_3}}{M_{Ne}}}$$

$$\Rightarrow \quad V_{SO_3} = \frac{V_{Ne} \times t_{SO_3}}{t_{Ne}} \times \sqrt{\frac{M_{Ne}}{M_{SO_3}}} = \frac{20 \times 30}{60} \times \sqrt{\frac{20}{80}} = \frac{10}{2} = 5\,dm^3$$

Ex.23 ***A gaseous mixture of O_2 and X containing 20% (mole %) of X, diffused through a small hole in 234 seconds while pure O_2 takes 224 seconds to diffuse through the same hole. Molecular weight of X is :***

Sol. $$\frac{t_{mix}}{t_{O_2}} = \sqrt{\frac{M_{mix}}{M_{O_2}}}$$

$$\frac{234}{224} = \sqrt{\frac{M_{mix}}{32}}$$

$\therefore \quad M_{mix} = 34.921.$

As the mixture contains 20% (mole %) of X, the molar ratio of O_2 and X may be represented as 0.8n : 0.2n, n being the total no. of moles.

$$\therefore \quad M_{mix} = \frac{32 \times 0.8n + M_x \times 0.2n}{n} = 34.921$$

$\therefore \quad M_x$ (mol. wt. of X) = 46.6

Ex.24 ***At 1200°C, mixture of Cl_2 and Cl atoms (both in gaseous state) effuses 1.16 times as fast as krypton effuses under identical conditions. Calculate the percentage of chlorine molecules dissociated into atoms. M (Kr) = 83.8 g mol^{-1}.***

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Sol. $Cl_2 \rightleftharpoons 2Cl$

$$\frac{r(Cl_2 \text{ and } Cl \text{ mix})}{r(Kr)} = 1.16\sqrt{\frac{M(Kr)}{M_{av}(Cl_2+Cl)}} = \sqrt{\frac{83.8}{M_{av}}}$$

$$\therefore \quad M_{av} = \frac{83.8}{(1.16)^2} = 62.28 \text{ g mol}^{-1}$$

	$Cl_2 \rightleftharpoons$	$2Cl$
Initial mole	1	0
After dissociation	(1 – x)	2x

(x = degree of dissociation)

Total moles after dissociation = 1 – x + 2x = (1 + x)

$$\therefore \quad \frac{(1-x)M(Cl_2)+2xM(Cl)}{(1+x)} = 62.28 \Rightarrow \frac{(1-x)\times 71+2x\times 35.5}{1+x} = 62.28$$

$x = 0.14$

$\therefore$ % dissociation = 14%

Ex.25 ***A mixture containing 2 moles of D_2 and 4 moles of H_2 is taken inside a container which is connected to another empty container through a nozzle. The nozzle is opened for certain time and then closed. The second bulb was found to contain 4 gm D_2. Then find % by moles of the lighter gases in second container.***

Sol. **I** **II**



lighter gas = H_2

$$\frac{r_{H_2}}{r_{D_2}} = \frac{n_{H_2}}{n_{D_2}}\sqrt{\frac{M_{D_2}}{M_{H_2}}} = \frac{n'_{H_2}/t}{n'_{D_2}/t}$$

$$(\%\text{ mole of } H_2)_{II} = \left(\frac{n'_{H_2}}{n'_{H_2}+n'_{D_2}}\right)\times 100$$

$$\therefore \quad \frac{n_{H_2}}{n_{D_2}}\sqrt{\frac{M_{D_2}}{M_{H_2}}} = \frac{n'_{H_2}}{n'_{D_2}} \quad \Rightarrow \quad \frac{4}{2}\sqrt{\frac{4}{2}} = \frac{n'_{H_2}}{1}$$

$$\Rightarrow \quad n'_{H_2} = 2\sqrt{2} \quad \Rightarrow \quad n'_{D_2} = 1$$

$$(\%\text{ mole of } H_2)_{II} = \frac{(2\sqrt{2})}{(2\sqrt{2}+1)}\times 100 = \frac{2\times 1.44}{[(2\times 1.44)+1]}\times 100 = \frac{(2.8)}{(2.8+1)}\times 100 = \frac{(2.8)}{(2.8+1)}\times 100$$

$= 73.87\%$

Ex.26 According to Graham's law , at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ ***of gases A and B is given by :*** **[JEE 1998]**

(A) $\frac{P_A}{P_B}\left(\frac{M_A}{M_B}\right)^{1/2}$ (B) $\left(\frac{M_A}{M_B}\right)\left(\frac{P_A}{P_B}\right)^{1/2}$ (C) $\frac{P_A}{P_B}\left(\frac{M_B}{M_A}\right)^{1/2}$ (D) $\frac{M_A}{M_B}\left(\frac{P_B}{P_A}\right)^{1/2}$

Ans. (C)

Sol. $r \propto \frac{P}{\sqrt{M}}$

$\therefore \frac{r_A}{r_B} = \frac{P_A}{P_B}.\sqrt{\frac{M_B}{M_A}}$

Ex.27 The density of the gaseous mixture in a vessel (CH_4 and He) at 2 atmosphere pressure and 300 K is 0.9756 g/lit. If a small pin-hole is made on the wall of the vessel, through which gases effuse, then which of the followings is the correct composition (by volume) of the gases CH_4 and He effusing out initially ?

(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 1 : 4

Ans. (A)

Sol. PM = dRT

$\therefore M_{avg} = \frac{dRT}{P} = \frac{0.9756 \times 0.082 \times 300}{2} \simeq 12$

Let the mixture contains x moles CH_4 and y moles He. Then

$\therefore M_{avg} = 12 = \frac{x \times 16 + y \times 4}{x + y} \Rightarrow \frac{x}{y} = \frac{2}{1}$

Now, $\frac{r_{CH_4}}{r_{He}} = \frac{P_{CH_4}}{P_{He}}.\sqrt{\frac{M_{He}}{M_{CH_4}}} = \frac{2}{1} \times \sqrt{\frac{4}{16}} = \frac{1}{1}$

Ex.28 The number of effusion steps required to convert a mixture of H_2 and O_2 from 240 : 1600 (by mass) to 3072 : 20 (by mass) is

(A) 2 (B) 4 (C) 5 (D) 6

Ans. (C)

Sol. $\left(\frac{n_{H_2}}{n_{O_2}}\right)_{final} = \left(\frac{n_{H_2}}{n_{O_2}}\right)_{initial} \times \left(\sqrt{\frac{M_{O_2}}{M_{H_2}}}\right)^n$

or $\frac{3072}{20} = \frac{240}{1600} \times \left(\sqrt{\frac{32}{2}}\right)^n \Rightarrow n = 5$

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9. KINETIC THEORY OF GASES

This is a theoretical model for ideal gas which can correlate the experimental facts (like Boyle's law, Charle's law& Avogadro's law etc.). It was presented by ***Bernoulli in 1738*** and developed in 1860 by ***Clausius, Maxwell, Kroning*** *and* ***Boltzmann***. Postulates of kinetic theory of gases are :

(i) All the gases consist of very small molecules or atoms whose volume is negligible compared to volume of container.

(ii) There are no attractive or repulsive forces between the molecules.

(iii) The gaseous molecules are under a continuous state of motion which is unaffected by gravity (the random straight line motion is known as brownian motion)

(iv) Due to the continuous motion, collision between gaseous molecules and with the wall of container occurs. The collision with the wall of container are responsible for pressure exerted by the gas on the wall of container.

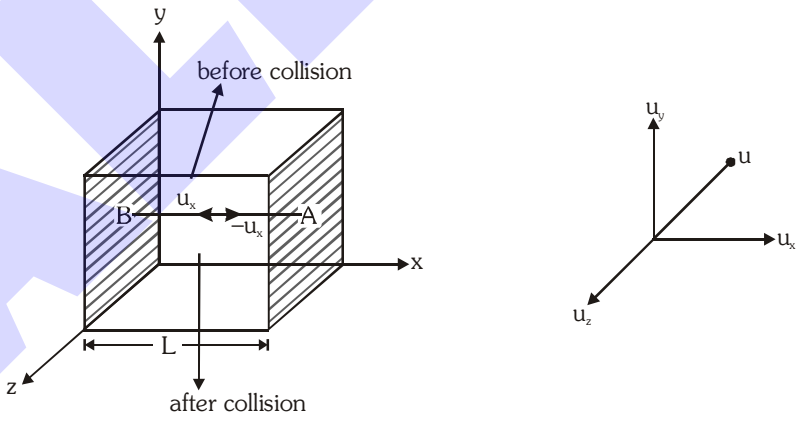
(v) The molecule moves with different speed, however the speed of each molecule keep on changing as the collision occur.

(vi) Collision among gas particles molecules is perfectly elastic, i.e., there is no loss in kinetic energy and moment during such collision.

(vii) The average kinetic energy of gas particles will depends on absolute temperature only.

9.1 Kinetic gas equation : Let us consider a cube of side length L, that has N molecules each of mass 'm' moving with velocity $\vec{u}$ in all direction and thus colliding with one each other and against sides of the container. Velocity $\vec{u}$ can be resolved into three components u_x, u_y and u_z along there axis such that

$$u^2 = u_x^2 + u_y^2 + u_z^2$$



For a simplest case we consider motion of a molecule along x-axis only in which it moves towards face B with velocity u_x. After collision against face B it moves towards face A with velocity $(-u_x)$, collision being elastic (which results in change in direction but not speed)

Hence, Momentum before collision on face B = mu_x

Momentum after collision on face B = $-mu_x$

Change in momentum due to one collision on face B = $mu_x - (-mu_x) = 2mu_x$

To strike face B again distance travelled = 2L

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Time taken to strike face B again = $\frac{2L}{u_x}$ seconds

$\therefore$ Number of collisions per second on face B along x-axis = $\frac{u_x}{2L}$

$\therefore$ Rate of change in momentum due to $\frac{u_x}{2L}$ collisions per second on face B along x-axis.

$= 2\,mu_x \cdot \frac{u_x}{2L} = \frac{mu_x^2}{L}$

Similarly for y-axis change in momentum per second = $\frac{mu_y^2}{L}$ and for z-axis = $\frac{mu_z^2}{L}$

Net force by N molecules on a wall , $F_x = \frac{mu_{x_1}^2}{L} + \frac{mu_{x_2}^2}{L} ..... + \frac{mu_{x_N}^2}{L} = \frac{M}{L}.\Sigma u_x^2$

Now pressure = $\frac{\text{Force}}{\text{Area of six faces}} = \frac{\frac{m}{L}.\Sigma u_x^2}{L^2} = \frac{m.\Sigma u_x^2}{L^3} = \frac{m.\Sigma u^2}{V}$ [L^3 = volume V]

$\therefore$ $PV = m.\Sigma u_x^{\,2}$

As $\Sigma u_x^2 = \Sigma u_y^2 = \Sigma u_z^2$ and $\Sigma u_x^2 + \Sigma u_y^2 + \Sigma u_z^2 = \Sigma u^2$

$\therefore$ $PV = m.\Sigma u_x^{\,2} = \frac{1}{3} m.\Sigma u^2$

$PV = \frac{1}{3} m \left(\frac{u_1^2 + u_1^2 + ...u_N^2}{N} \right).N$

$\therefore$ $\boxed{PV = \frac{1}{3} mNu_{rms}^2}$

This equation is called kinetic gas equation.

9.2 Kinetic energy of gas molecules :

Total translational K.E. of molecules

$= \frac{1}{2}mu_2^2 + \frac{1}{2}mu_2^2 + ... + \frac{1}{2}m.u_N^{\,2} = \frac{1}{2}m.\Sigma u^2 = \frac{1}{2}\,mN.u_{rms}^2 = \frac{3}{2}PV = \frac{3}{2}nRT$

$\therefore$ Average translational K.E. per mole $= \frac{3}{2}RT$

and $(K.E.)_{\text{per molecule}} = \frac{3}{2}\left(\frac{R}{N_A}\right)T = \frac{3}{2}kT$

Where k = Boltzman constant = $\frac{R}{N_A} = \frac{8.314 J/mol K}{6.02 \times 10^{23}} = 1.3806 \times 10^{-23}\ J\ K^{-1}$

Note : $(K.E.)_{\text{per molecule}}$ and $(K.E.)_{\text{per mol}}$ is only depend on absolute temperature. It is does not depend on the nature of gas. This conclusion is known as ***"Maxwell's Generalisation"***.

9.3 Root Mean Square Velocity (u_{rms}) by kinetic gas equation :

$PV = \frac{1}{3} mN\, u_{rms}^2$

$\therefore$ $u_{rms} = \sqrt{\frac{3PV}{mN}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3RT}{M}}$

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9.4 Diffrent kind of speed of molecules :

(i) Average or mean speed, $u_{av} = \frac{u_1 + u_2 + ... + u_N}{N} = \sqrt{\frac{8RT}{\pi M}}$

(ii) Root mean square speed, $u_{rms} = \sqrt{\frac{u_1^2 + u_2^2 + ... + u_N^2}{N}} = \sqrt{\frac{3RT}{M}}$

(iii) Most probable speed, $u_{mp} = \sqrt{\frac{2RT}{M}}$

It is the speed at which maximum fraction of molecules are travelling

9.5 Ratio of speeds :

$$U_{rms} : U_{avg} : U_{mps} = \sqrt{\frac{3RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{2RT}{M}}$$

$$= \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} = 1.225 : 1.128 : 1.000 = 1.00 : 0.92 : 0.816$$

Ex.29 Four particles have speed 2, 3, 4 and 5 cm/s respectively. Find their average & rms speed.

Sol. $U_{avg.} = \frac{U_1 + U_2 + U_3 + ...U_N}{N} = \frac{2+3+4+5}{4} = 3.5\,cm/s$

$$U_{r.m.s.} = \sqrt{\frac{U_1^2 + U_2^2 + U_3^2 + ...}{N}} = \sqrt{\frac{2^2 + 3^2 + 4^2 + 5^2}{4}} = 3.67 \text{ cm/s.}$$

Ex.30 The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : **[JEE 2000]**

(A) $T(H_2) = T(N_2)$ (B) $T(H_2) > T(N_2)$

(C) $T(H_2) < T(N_2)$ (D) $T(H_2) = \sqrt{7}\ T(N_2)$

Ans. (C)

Sol. $(u_{rms})_{H_2} = \sqrt{7} \times (u_{rms})_{N_2}$

$$\sqrt{\frac{3RT_1}{2}} = \sqrt{7} \times \sqrt{\frac{3RT_2}{28}}$$

$$\frac{3RT_1}{2} = \frac{7 \times 3RT_2}{28}$$

$2T_1 = T_2 \quad \Rightarrow \quad T_{N_2} > T_{H_2}$

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Ex.31 ***At what temperature do the average speed of $CH_{4(g)}$ molecule equal the average speed of O_2 molecule at 300 K ?***

Sol. $(U_{avg})_{CH_4} = (U_{avg})_{O_2}$

$$\sqrt{\frac{8RT}{\pi \times 16}} = \sqrt{\frac{8 \times R \times 300}{\pi \times 32}}$$

$T = 150$ K

Ex.32 ***At 27°C, find the ratio of root mean square speeds of ozone to oxygen.***

Sol. $$\frac{U_{rms}(O_3)}{U_{rms}(O_2)} = \sqrt{\frac{\frac{3RT}{M.W_{O_3}}}{\frac{3RT}{M.W_{O_2}}}} = \sqrt{\frac{M.W_{O_2}}{M.W_{O_3}}} = \sqrt{\frac{32}{48}} = \sqrt{\frac{2}{3}}$$

Ex.33 ***The temperature at which U_{rms} of He becomes equal to U_{mp} of CH_4 at 500 K.***

Sol. $(U_{rms})_{He} = (U_{mp})_{CH_4}$

$$\sqrt{\frac{3RT_{He}}{M_{He}}} = \sqrt{\frac{2RT_{CH_4}}{M_{CH_4}}}$$

$$\frac{3T}{4} = \frac{2 \times 500}{16}$$

$$T = \frac{250}{3} K$$

Ex.34 ***Calculate the root mean square speed of H_2 molecules under following condition.***

(a) 2 mole of H_2 at 27°C.

(b) 3 mole of H_2 in a 5 lit container at 10^5 Pa.

(c) 4 mole of H_2 at the density of 1 gm/ml at 10^5 Pa.

Sol. (a) $$U_{r.m.s.} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 J/mol K \times 300 K}{2 \times 10^{-3} kg}} = 1934.24 \text{ m/sec.}$$

(b) $$U_{r.m.s.} = \sqrt{\frac{3PV}{nM}} = \sqrt{\frac{3 \times 10^5 Pa \times 5 \times 10^{-3} m^3}{3 mol \times 2 \times 10^{-3} kg}} = 500 \text{ m/sec.}$$

(c) $$U_{r.m.s.} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3 \times 10^5 Pa}{10^3 kg/m^3}} = 17.32 \text{ m/sec.}$$

Ex.35 ***Calculate the total translational kinetic energy of 8 gram methane (CH_4) at 27°C temperature.***

Sol. $n = \frac{8}{16} = \frac{1}{2}$, $T = (27° + 273) = 300$ K, $R = 8.314$ J mol^{-1} K^{-1}

$$(K.E.)_{n \text{ mol}} = n \times \frac{3}{2}RT = \frac{1}{2} \times \frac{3}{2} \times 8.314 \times 300 = 1870.65 \text{ J}$$

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Ex.36 Calculate the pressure exerted by 10^{23} gas molecules, each of mass 10^{-25} kg, in a container of volume 1×10^{-3} m^3 and having root mean square velocity of 10^3 ms^{-1}. Also calculate total kinetic energy and Temperature of the gas.

Sol. By kinetic theory

$$P = \frac{1}{3}\frac{mNu^2}{V} = \frac{1\times10^{-25}\times10^{23}\times(10^3)^2}{3\times10^{-3}} = 3.33 \times 10^6 \text{ N m}^{-2}$$

$$\text{Total KE} = \left(\frac{1}{2}mu_{rms}^2\right) \times \text{ N} = \frac{1}{2}\times 10^{-25}\times(10^3)^2\times10^{23} = \frac{1}{2}\times10^4 = 0.5\times10^4 \text{ J}$$

$$\text{Also total KE} = \frac{3}{2}nRT, \text{ where n (mole)} = \frac{10^{23}}{N_A} = \frac{10^{23}}{6.023\times10^{23}}$$

$$0.5\times10^4 = \frac{3}{2}\times\frac{10^{23}}{6.023\times10^{23}}\times8.314\times T$$

$$\therefore \quad T = \frac{0.5\times10^4\times2\times6.023}{3\times8.314} = 2415 \text{ K}$$

Ex.37 At what temperature will be total kinetic energy (KE) of 0.30 mole of He be the same as the total KE of 0.40 mole of Ar at 400 K :

(A) 400 K (B) 373 K (C) 533 K (D) 300 K

Ans. (C)

Sol. Total K.E. = $\frac{3}{2}nRT$

From question, $\frac{3}{2}\times0.3\times R\times T = \frac{3}{2}\times0.4\times R\times400$

$\therefore$ T = 533.33 K.

10. MAXWELL'S DISTRIBUTION OF SPEEDS

It has already been pointed out that a gas is a collection of tiny particles separated from one another by large empty spaces and moving rapidly at random in all directions. In the course of their motion, they collide with one another and also with the walls of the container. Due to frequent collisions, the speeds and directions of motion of the molecules keep on changing. Thus, all the molecules in a sample of gas do not have same speed. Although it is not possible to find out the speeds of individual molecule , yet from probability considerations it has become possible to work out the distribution of molecules in different speed intervals. This distribution is referred to as the Maxwell-Boltzmann distribution in honour of the scientists who developed it. It may be noted that the distribution of speeds remains constant at a particular temperature although individual speeds of molecules may change.

$$dN = 4\pi N\left[\frac{M}{2\pi RT}\right]^{\frac{3}{2}} e^{-\frac{Mu^2}{2RT}}\,u^2du = 4\pi N\left[\frac{m}{2\pi kT}\right]^{3/2} e^{\frac{-mu^2}{2KT}}.u^2.du$$

Here, dN = Number of molecules having speeds between u and u + du.

N = Total number of molecules.

M = Molar mass of gas

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dN / N = fraction of molecules having speeds between u and u + du.

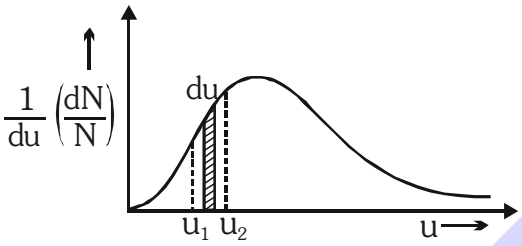
and $\frac{1}{N}\left(\frac{dN}{du}\right)$ = fraction of molecules having speed between u to u + du per unit interval of speed.

= Maxwell distribution function = f(u)

According to this expression, the fraction $\left(\frac{dN}{N}\right)$ of molecules depends only on temperature having speeds between u and u + du for a gas of molar mass M. Thus for a given temperature, this fraction has a constant value.

10.1 Properties of Maxwell's graph :

10.1.1

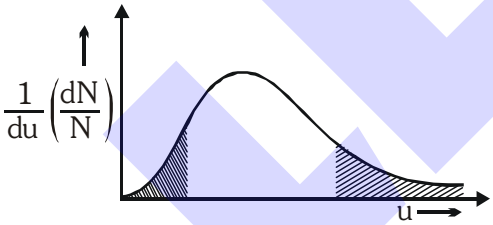


Area between u_1 and $u_2 = \int_{u_1}^{u_2} \frac{1}{du}\left(\frac{dN}{N}\right)du = \int_{u_1}^{u_2}\left(\frac{dN}{N}\right)$

Hence, total fraction of particles with speed between u_1 and u_2

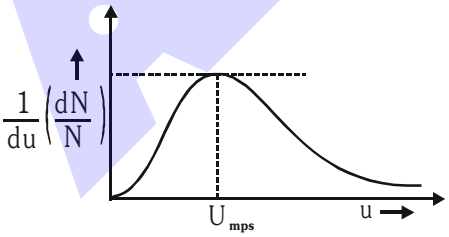
= Area under the curve represents fraction of molecules.

10.1.2



It can be seen from the above figure, that the fraction of molecules having either very low speeds or very high speeds are small in numbers.

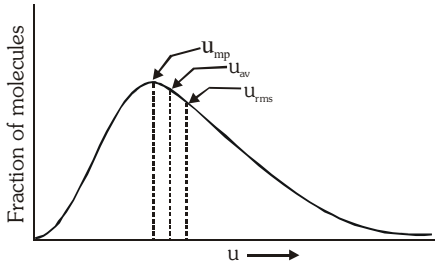
10.1.3



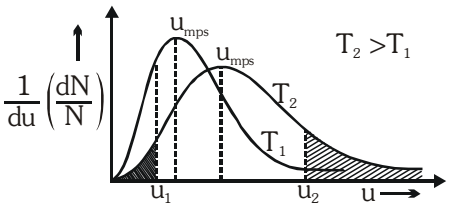
The curve at any temperature is parabolic near the origin, since the factor u^2 is dominant in this region, the exponential function being approximately equal to unity. At high values of u, however, the exponential factor dominates the behaviour of the function, causing it to decrease rapidly in value. As a consequence of the contrasting behaviour of two factors, the product function passes through a maximum at a speed known as the most probable speed (u_{mps}). Thus, the most probable speed is the speed possessed by the maximum fraction of the molecules.

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10.1.4 Graph between fraction of molecules vs molecular speeds :



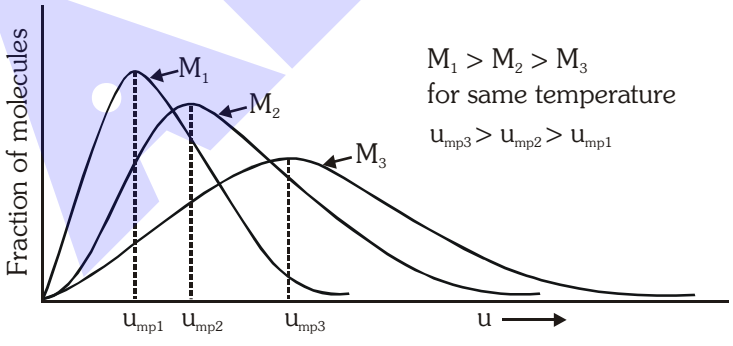
10.1.5



Total area under the curve will be constant and will be unity at all temperatures. The above figure illustrates the distribution of speeds at two temperatures T_1 and T_2. Since the total no. of molecules is same at both temperatures, increase in the K.E. of the molecules results decrease in fraction of molecules having lower speed range and increase in fraction of molecules having higher speed range on increasing the temperature.

On increasing temperature the value of u_{mps}(most probable speed) will increase. Also the curve at the higher temperature T_2 has its u_{mps} shifted to a higher value compared with that for T_1, whereas corresponding fraction of molecules has decreased. But at the same time, the curve near u_{mps} has become broader at the higher temperature indicating the more molecules possess speeds near to most probable speed.

10.1.6



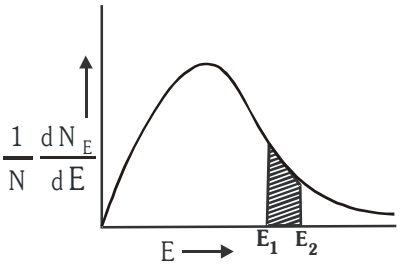
At a given temperature u_{mps} will be more for lighter gas (M_3) but fraction of molecules moving with u_{mps} will be more for heavier gas (M_1).

Note : Effect of M and T are opposite.

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10.2 Maxwell Distribution of kinetic energy :

$$dN = 4\pi N\left(\frac{M}{2\pi RT}\right)^{3/2} .e^{\frac{-Mu^2}{2RT}}\ u^2.du = 2\pi\left(\frac{1}{\pi RT}\right)^{3/2}\sqrt{E}.e^{-E/RT}.dE$$



The shaded area of this graph indicate the fraction of particles having energy between E_1 and E_2

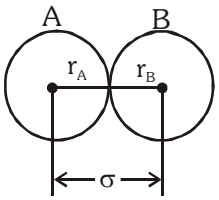
11. COLLISION PARAMETERS :

Assumption :

All the particles (molecules or atoms) are rigid , hard spheres and their in nature do not change after collision.

11.1 Collision diameter (σ) :

It is the closest distance between the centres of two molecules taking part in collision.



Collision diameter $(\sigma) = r_A + r_B$

11.2 Collision Number (z_1) :

The no. of collisions made by a single molecule with other molecules per unit time is called **(collision number)**.

$$Z_1 = \sqrt{2}\,\pi\sigma^2\,U_{avg}N^*$$

11.3 Collision Frequency :

It is the total number of biomolecular molecular collisions taking place per unit time per unit volume of the gas.

The total number of bimolecular collision per unit time per unit volume between similar particles is given as

$$Z_{11} = \frac{1}{2}(Z_1N^*) = \frac{1}{2}\times N^* \times \sqrt{2}\,\pi\sigma^2\,U_{avg}N^*$$

$$Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2\,U_{avg}N^{*2}$$

If the collisions involve two unlike molecules, the no. of bimolecular collision per unit time per unit volume is given as

$$Z_{12} = \pi\sigma_{12}^2\left(\frac{8kT}{\pi\mu}\right)^{\frac{1}{2}} N_1^*N_2^*$$

Where N_1^* and N_2^* are the no. of molecules per unit volume of the two types of gases, σ_{12} is the average diameter of two molecules, that is $\sigma_{12}=\frac{\sigma_1+\sigma_2}{2}$ & μ is the reduced mass, that is $\mu=\frac{m_1m_2}{(m_1+m_2)}$, m_1 & m_2 are the mass of single molecule respectively 1 and 2.

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11.4 Mean free path :

The mean free path is the average distance travelled by a molecule between two successive collisions. We can express it as follows :

$$\lambda = \frac{\text{distance travelled per unit time}}{\text{No. of collsions made by single molecule per unit time}}$$

$$= \frac{U_{avg}}{Z_1} = \frac{U_{avg}}{\sqrt{2}\pi\sigma^2 U_{avg} N^*} = \frac{1}{\sqrt{2}\pi\sigma^2 N^*}$$

11.5 Wall collision : It represents the total number of molecules colliding at the wall per unit area per unit time.

$$Z_w = \frac{1}{4}.N^*.u_{av} = \frac{P.N_A}{\sqrt{2\pi MRT}}$$

11.6 Rate of effusion :

If the corss-section area of orifice in the vessel is 'A', then the number of molecules effusing out per unit time is

$$r_{eff.} = Z_w.A = \frac{P.N_A.A}{\sqrt{2\pi MRT}}$$

Ex.38 Calculate λ, Z_1 and Z_{11} for oxygen at 298 K and 10^{-3} mm Hg. Given $\sigma = 3.61 \times 10^{-8}$ cm.

Sol. $N^* = \frac{P}{kT} = \frac{10^{-3} \times 101325}{760 \times 1.38 \times 10^{-23} \times 298} = 0.324 \times 10^{20}$

$$U_{avg} = \sqrt{\frac{8}{\pi}\frac{RT}{M}} = \sqrt{\frac{8}{3.14} \times \frac{8.314 \times 298}{32 \times 10^{-3}}} = 444.138 \text{ m/sec}$$

$$Z_1 = \sqrt{2}\,\pi\,\sigma^2\,U_{avg}\,N^* = \sqrt{2} \times 3.14 \times 3.61 \times 10^{-10}\,\text{m} \times 444.138\,\text{m} \times 0.324 \times 10^{20}$$

$$= 8.326 \times 10^3 \text{ sec}^{-1}$$

$$Z_{11} = \frac{1}{2} Z_1 N^* = \frac{1}{2} \times 8.326 \times 10^3 \text{ sec}^{-1} \times 0.324 \times 10^{20} = 13.488 \times 10^{22} \text{ m}^{-3} \text{ sec}^{-1};$$

$$\lambda = \frac{U_{avg.}}{Z_1} = \frac{444.138 \text{ m/sec}}{8.326 \times 10^3 \text{ sec}^{-1}} = 5.334 \times 10^{-2} \text{ m}$$

Ex.39 Two flask A and B have equal volume. A is maintained at 300 K and B at 600 K while A contains H_2 gas, B has an equal mass of CH_4 gas. Assuming ideal behaviour for both the gases, find the following.

(a) Flask containing greater number of moles

(b) Flask in which pressure is greater

(c) Flask in which U_{avg} of the molecules are greater

(d) Flask with greater mean free path of molecules
(Collision diameters of H_2 & CH_4 may be taken same)

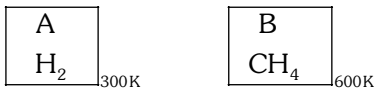
(e) Flask with greater molar kinetic energy.

(f) Flask in which the total kinetic energy is greater

(g) Flask in which Z_1 and Z_{11} are greater

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Sol.



(a) $N_{H_2} = \frac{m}{2}N_A$; $N_{CH_4} = \frac{m}{16}N_A$

$\therefore$ molecules of H_2 in flask A > molecules of CH_4 in flask B

(b) $P_AV = n_ART_A$; $P_BV = n_BRT_B$

$\Rightarrow \frac{P_A}{P_B} = \frac{n_AT_A}{n_BT_B} = \frac{m/2}{m/16} \times \frac{300}{600} = 4$

$\therefore$ pressure of H_2 in flask A > pressure of CH_4 in flask B

(c) $(U_{avg})_A = \sqrt{\frac{8}{\pi}\frac{RT_A}{M_A}}$; $(U_{avg})_B = \sqrt{\frac{8}{\pi}\frac{RT_B}{M_B}}$

$\frac{(U_{avg})_A}{(U_{avg})_B} = \sqrt{\frac{T_A}{T_B} \times \frac{M_B}{M_A}} = \sqrt{\frac{300}{600} \times \frac{16}{2}} = 2$

$\therefore$ U_{avg} of H_2 in flask A > U_{avg} of CH_4 in flask B

(d) $\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 N^*} = \frac{1}{\sqrt{2}\pi\sigma^2 \times P} \times kT$ [where N* = P/kT]

$\frac{\lambda_A}{\lambda_B} = \frac{T_A}{T_B} \times \frac{P_B}{P_A} = \frac{300}{600} \times \frac{1}{4} = \frac{1}{8}$

$\therefore$ $\lambda_B > \lambda_A$

(e) molar K.E. $= \frac{3}{2}RT$

$\therefore$ $T_B > T_A$

$\therefore$ KE of CH_4 in flask B > KE of H_2 in flask A

(f) $(KE)_{total} = \frac{3}{2}RT \times n$

$\frac{(KE)_{T,A}}{(KE)_{T,B}} = \frac{(300 \times \frac{m}{2})}{(600 \times \frac{m}{16})} = \frac{1}{2} \times 8 = 4$

$\therefore$ $(KE)_{T,A} > (KE)_{T,B}$

(g) (i) $Z_1 = \sqrt{2}\pi\sigma^2 U_{avg} N^*$

$\frac{(Z_1)_A}{(Z_1)_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{N_A^*}{N_B^*} = 2 \times \frac{N_A}{N_B} = 2 \times 8 = 16$

$\therefore$ $(Z_1)_A > (Z_1)_B$ $(Z_1)_A > (Z_1)_B$

(ii) $Z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^{*2}$

$\frac{(Z_{11})_A}{(Z_{11})_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{(N_A^*)^2}{(N_B^*)^2} = 2 \times \frac{N_A}{N_B} = 2 \times 8^2 = 128$

$\therefore$ $(Z_{11})_A > (Z_{11})_B$

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SELF STUDY

12. PRESSURE MEASURING DEVICES :

Generally, the instruments used for the measurement of pressure of a gas are ***barometer*** and ***manometer***, which measures the pressure in terms of height of mercury column.

$$Pressure(P) = Force(F) / Area(A)$$

$$= Mass(m) \times Acceleration(g) / Area(a)$$

$$= Volume \times density \times Acceleration(g) / Area(a)$$

$$= Area\ (a) \times height\ (h) \times density\ (\rho) \times Acceleration(g) / Area(a)$$

$$Pressure(P) = h\rho g$$

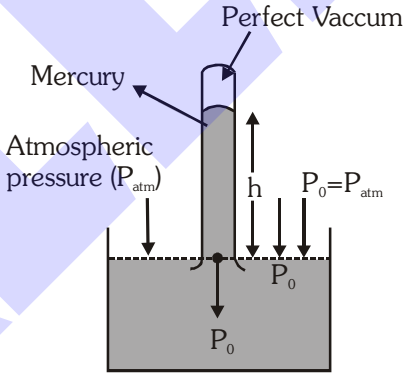
where h = Height of mercury column in the barometer.

ρ = Density of mercury.

g = Acceleration due to gravity.

Pressure does not depend on the cross section of tube, but only on the vertical height of the Hg. If area is doubled, volume also gets doubled and mass will also gets doubled. Now it will rest on twice area but pressure exerted remains same.

(i) Barometer : A barometer is an instrument that is used for the measurement of atmospheric pressure. The construction of the barometer is as follows -



A thin narrow calibrated capillary tube is filled up to the brim, with a liquid such as mercury, and is inverted into a trough filled with the same fluid. Now depending on the external atmospheric pressure, the level of the mercury inside the tube will adjust itself, the reading of which can be monitored. When the mercury column inside the capillary comes to rest, then the net forces on the column should be balanced.

$\Rightarrow \quad P_0 \times A = \rho A \times gh$

$\Rightarrow \quad \boxed{P_0 = \rho gh}$; where ρ is the density of the fluid.

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(ii) Manometer :

(a) Open end manometer :

It consists of a U-shaped tube partially filled with mercury. One limb of the tube is shorter than the other. The shorter limb is connected to the vessel containing the gas whereas the longer limb is open as shown in fig. The mercury in the longer tube is subjected to the atmospheric pressure while mercury in the shorter tube is subjected to the pressure of the gas.

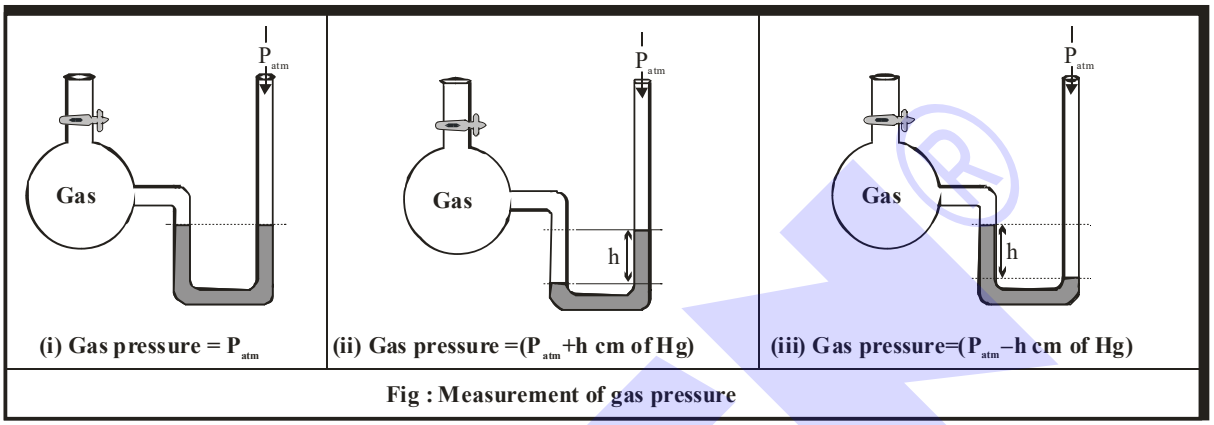


Fig : Measurement of gas pressure

Where P_{atm} = 76 cm of Hg and h = Height in cm of Hg

There are three possibilities as described below :

(i) If the level of Hg in the two limbs is same, then gas pressure = atmospheric pressure (P_{atm}).

(ii) If the level of Hg in the longer limbs is higher, gas pressure

$= P_{atm}$ + (difference between the two levels)

$= P_{atm}$ + h cm of Hg

(iii) If the level of Hg in the shorter limb is higher, then gas pressure

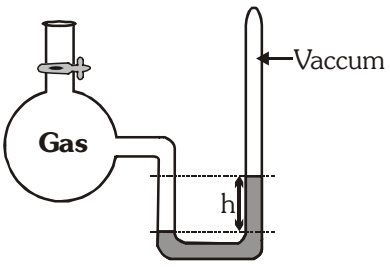
$= P_{atm}$ – (difference between the two levels)

$= P_{atm}$ – h cm of Hg

(b) Closed end manometer :

This is generally used to measure low gas pressure.

It also consists of U-tube with one limb shorter than the other and partially filled with mercury as shown in fig. The space above mercury on the closed end is completely evacuated. The shorter limb is connected to the vessel containing gas. The gas exerts pressure on the mercury in the shorter limb and forces its level down.



Closed end manometer

Gas pressure = [Difference in the Hg level in two limbs]

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Ex.40 Why mercury is used in the barometer tube?

Sol. Mercury, a liquid with very high density, is normally used in the barometer because it does not stick to the surface of the glass tube. Mercury is also non-volatile at room temperature. Therefore, there are hardly any vapours of mercury above the liquid column and their pressure, if any , can be neglected. Due to high density of mercury, height of mercury column will be small and can be easily measured.

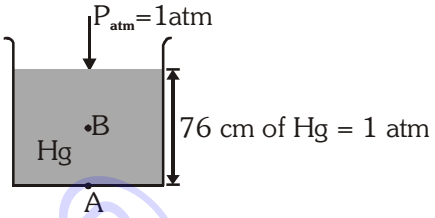
Ex.41 An open tank is filled with Hg upto a height of 76cm.

Find the pressure at the

(a) Bottom (A) of the tank

(b) Middle (B) of the tank.

(If atmospheric pressure is 1 atm)



Sol. **(a)** At bottom,

$$P_A = P_{atm} + P_{Hg}$$

$$= 1 + 1 = 2 \text{ atm}$$

(b) At middle,

$$P_B = P_{atm} + P_{Hg}$$

$$= 1 + \frac{1}{2} = 1.5 \text{ atm}$$

Ex.42 Find the height of water upto which water must be filled to create the same pressure at the bottom, as in above problem.

Given that $d_w = 1$ gm/cm^3, $d_{Hg} = 13.6$ gm/cm^3, $h_{Hg} = 76$ cm

Sol. $P_{water} = P_{Hg}$

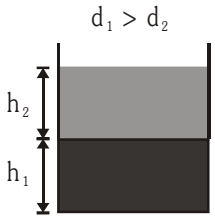
$h_w d_w g = h_{Hg} d_{Hg} g$

$h_w d_w = h_{Hg} d_{Hg}$

$h_w \times 1 \text{ g/cm}^3 = 76 \text{ cm} \times 13.6 \text{ g/cm}^3$

$h_w = 1033.6 \text{ cm} = 10.336 \text{ m}.$

Ex.43 What will be the pressure if two immiscible fluid is filled according to given diagram.



(a) Find the pressure at the bottom of tank.

(b) Find the pressure at the middle point of bottom layer.

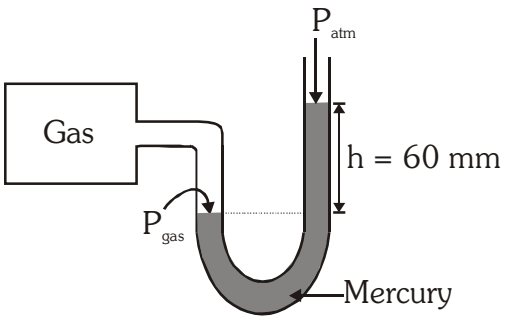
Sol. (a) $P_{atm} + h_2 d_2 g + h_1 d_1 g$;

(b) $P_{atm} + h_2 d_2 g + \frac{h_1}{2} d_1 g$

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Ex.44 Find the pressure of the gas inside a container if the open manometer attached to the container shows a difference of 60 mm.

Sol. **Case-I :** $P_{atm} < P_{gas}$

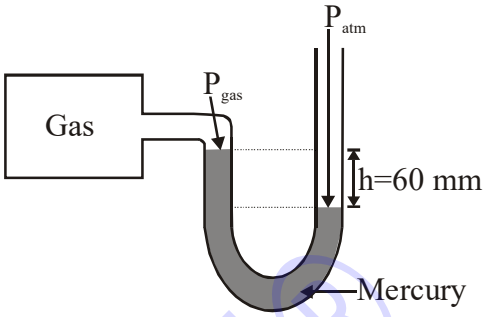


$P_{gas} = P_{atm} + 60$ mm

$= 760$ mm $+ 60$ mm

$= 820$ mm of Hg

Case-II : $P_{atm} > P_{gas}$



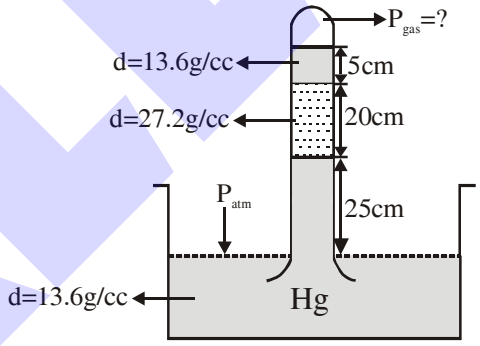
$P_{atm} = P_{gas} + 60$ mm

760 mm $= P_{gas} + 60$

$P_{gas} = 700$ mm

Ex.45 In the following arrangement find the pressure of gas (in cm of Hg).

(Assume that atmospheric pressure P_{atm} = 75 cm of Hg)



Ans.(5)

Sol. $P_{atm} = P_{gas} + P_1 + P_2 + P_3$

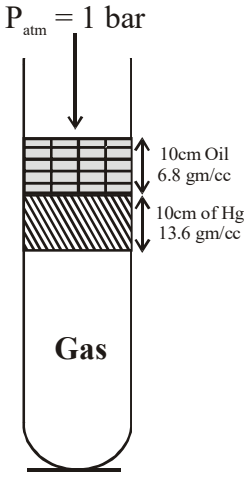
or 75 cm Hg = P_{gas} + 5 cm of liquid (1) + 20 cm of liquid (2) + 25 cm of Hg

$$= P_{gas} + \frac{5 \times 13.6}{13.6} \text{ cm Hg} + \frac{20 \times 27.2}{13.6} \text{ cm Hg} + 25 \text{ cm Hg}$$

$\therefore P_{gas} = 5$ cm Hg

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Ex.46



Find pressure of gas (in cm of Hg) enclosed in tube.

Ans. (90)

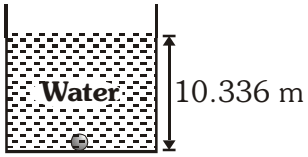
Sol. P_{gas} = 1 bar + 10 cm oil + 10 cm Hg

$$= 75 \text{ cm Hg} + \frac{10 \times 6.8}{13.6} \text{ cm Hg} + 10 \text{ cm Hg}$$

$$= 90 \text{ cm Hg.}$$

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***Ex.*47.(a)** Radius of a bubble at the bottom of the tank shown below was found to be 1 cm, then find the radius (in cm) of the bubble at the surface of water considering the temperature at the surface & bottom being same. **[$(1.26)^3 = 2.0$, $(1.14)^3 = 1.5$]**

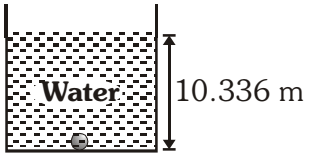


(b) If absolute temperature at the surface is 4 times that at the bottom, then find radius (in cm) of bubble at the surface.

IG0009

Ans. a. (1.26) b. (2.00)

Sol. (a)



Pressure of 10.336 m water = 1 atm

R_B = Radius of bubble at bottom = 1 cm

R_S = Radius of bubble at surface

Apply PV = nRT at bottom

$P_B V_B = nRT$

$$2 \times \frac{4}{3}\pi (R_B)^3 = nRT \quad \text{..... (1)}$$

Apply PV = nRT at surface $P_S.V_S = nRT$

$$1 \times \frac{4}{3}\pi (r_s)^3 = nRT \quad \text{..... (2)}$$

Diving (1) / (2)

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$$\Rightarrow \frac{2}{7}\times\left(\frac{R_B}{R_S}\right)^3 = 1$$

$$\Rightarrow 2 = \left(\frac{R_S}{R_B}\right)^3$$

$\because$ $R_B = 1$ cm

$R_S = 2^{1/3}$ cm

(b) $T_S = 4T_B$

T_S = temperature at surface

T_B = temperature at bottom

Apply PV = nRT at bottom

$P_B \times V_B = nRT_B$ ..... (1)

$$2\times\frac{4}{3}\pi(r_B)^3 = nRT_B$$

Apply PV = nRT at surface

$$1\times\frac{4}{3}\pi(r_S)^3 = nRT_S \quad \text{..... (2)}$$

On dividing equation 2 by 1

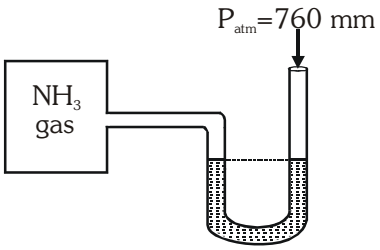
$$\frac{2\times\frac{4}{3}\pi(r_B)^3}{1\times\frac{4}{3}\pi(r_S)^3} = \frac{nRT_B}{nR(4T_B)}$$

$$2\times\left(\frac{r_B}{r_S}\right)^3 = \frac{1}{4}$$

$r_S = 2r_B = 2$cm

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***Ex.*48.** A manometer attached to a flask contains NH_3 gas have no difference in mercury level initially as shown in diagram. After the sparking into the flask, it have difference of 19 cm in mercury level in two columns. Calculate % dissociation of ammonia.



IG0026

Ans. (25.00)

Sol. $2NH_3 \longrightarrow N_2 + 3H_2$

t = 0	76 cm	0	0
$t = t_{eq}$	76–x	$\frac{x}{2}$	$\frac{3x}{2}$

Total pressure of system = $(76 - x) + \frac{x}{2} + \frac{3x}{2} = (76 + x)$ cm of Hg

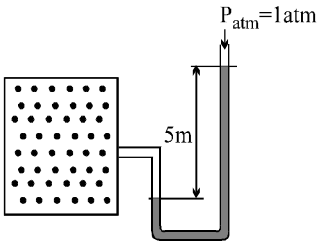
$76 + x = 95$ [$P_{total} = 76 + 19 = 95$]

$x = 19$ cm of Hg

% dissociation of $NH_3 = \frac{19}{76} \times 100 = 25\%$

***Ex.*49** Calculate the number of moles of gas present in the container of volume 10 lit at 300 K. If the manometer containing glycerin shows 5m difference in level as shown in diagram.

Given: $d_{glycerin}$ = 2.72 gm/ml, $d_{mercury}$ = 13.6 gm/ml.



IG0046

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Ans. (0.94)

Sol. Height of glycerin column = 5m = 500 cm

For glycerin

$(h\rho g)_{glycerin} = (h\rho g)_{Hg}$

$500 \times 2.72 \times g = h \times 13.6 \times g$

$h_{Hg} = 100$ cm

Hence $P_{gas} = P_{atm} + P_{glycerin}$

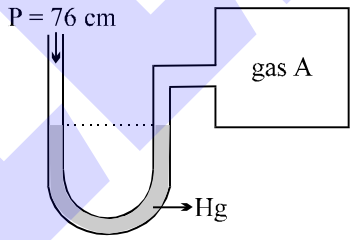
(Initial) = 76 + 100 = 176 cm of Hg

PV = nRT

$$\left(\frac{176}{76}\right) \times 10 = n \times 0.082 \times 300$$

Moles of gas (n) = 0.94 mole

Ex.50 An open ended mercury manometer is used to measure the pressure exerted by a trapped gas as shown in the figure. Initially manometer shows no difference in mercury level in both columns as shown in diagram.



After sparking 'A' dissociates according to following reaction

$$2A(g) \longrightarrow 3B(g) + 2C(g)$$

If pressure of Gas "A" decreases to 0.8 atm. Then (Assume temperature to be constant and is 300 K)

(A) total pressure increased by 1.3 atm

(B) total pressure increased by 0.3 atm

(C) total pressure increased by 22.3 cm of Hg

(D) difference in mercury level is 228 mm.

IG0128

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Ans. (B,D)

Sol. $2A(g) \longrightarrow 3B(g) + 2C(g)$

t = 0	1 atm	0	0
t	0.8 atm	0.3 atm	0.2 atm

$P_{total} = P_A + P_B + P_C$

$0.8 + 0.3 + 0.2 = 1.3$ atm.

* Total pressure is increased by 0.3 atm or 22.8 cm of Hg or 228 mm of Hg.

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SOLVED EXAMPLES

1. ***An evacuated glass vessel weighs 50.0 g when empty, 148.0 gm when filled with a liquid of density 0.98 g /mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 k . Determine the molecular weight of the gas .*** **[JEE 1998]**

Ans. 123

Sol. Weight of liquid = 148 – 50 = 98 g

$$\text{Volume of liquid} = \frac{98}{0.98} = 100\text{mL} = \text{volume of vessel}$$

Thus, vessel of 100 ml contains ideal gas at 760 mm of Hg at 300K.

Weight of gas = 50.5 – 50 = 0.5 g

Using PV = nRT

$$\frac{760}{760} \times \frac{100}{1000} \times \frac{0.5}{m} = 0.0821 \times 300$$

$\therefore$ Molecular weight of gas (m) = 123

2. ***The pressure exerted by 12 g of an ideal gas at temperature t ºC in a vessel of volume V is one atmp. When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10 %. Calculate the temperature 't' and volume 'V'.*** **[Molecular weight of gas = 120]** **[JEE 1999]**

Ans. –173°C , 0.82 L

Sol. $P_1 = 1 = 1$ atm, $T_1 = t + 273$

$$P_2 = 1 + \frac{10}{100} = 1.1 \text{ atm}, \; T_2 = t + 283$$

$$\frac{1 \times V}{1.1 \times V} = \frac{\frac{12}{M} \times R(t + 273)}{\frac{12}{M} \times R(t + 283)}$$

$t = -173^\circ C = 100K$

$$1 \times V = \frac{12}{120} \times 0.082 \times 100$$

$V = 0.82$ lite

3. ***One mole of N_2 gas at 0.8 atm takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atm takes 57 sec to diffuse through the same hole. Calculate the molecular formula of the compound.*** **(At. wt. Xe = 138, F = 19)** **[JEE 1999]**

Ans. XeF_6

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Sol. $\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$

$$\frac{n_1}{t_1} \times \frac{t_2}{n_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$$

$$\frac{1}{38} \times \frac{57}{1} = \frac{0.8}{1.6}\sqrt{\frac{M}{28}}$$

$M = 252$

$138 + 19 \times n = 252 \Rightarrow n = 6 \Rightarrow XeF_6$

4. The root mean square velocity of an ideal gas at constant pressure varies with density as

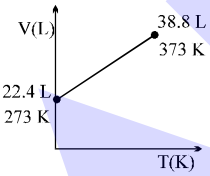
(A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$ **[JEE 2001]**

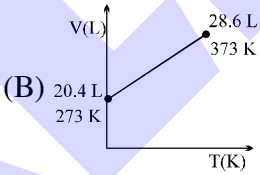
Ans. (D)

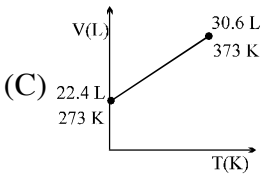
Sol. $u_{rms} = \sqrt{\frac{3P}{d}}$

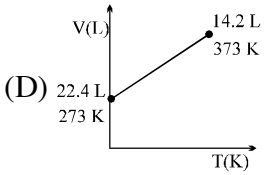
$$u_{rms} \propto \frac{1}{d^{1/2}}$$

5. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atmp? **[JEE 2002]**

(A) 

(B) 

(C) 

(D) 

Ans. (C)

Sol. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$$\frac{22.4}{273} = \frac{30.6}{373}$$

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6. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature. **[JEE 2003]**

Ans. 434.17 m/sec

Sol. $u_{rms} = \sqrt{\frac{3\pi}{8}} \times u_{avg}$

$= \sqrt{\frac{3 \times 3.14}{8}} \times 400 = 434\,m/s$

7. What volume of hydrogen gas, at 273K and 1 atm pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ? **[AIEEE-2003]**

(A) 44.8 L (B) 22.4 L (C) 89.6 L (D) 67.2 L

Ans. (D)

Sol. $2BCl_3 + 3H_2 \longrightarrow 2B + 6HCl$

3 mole 2 mole

$= 3 \times 22.4$ $= 2 \times 10.8 = 21.6$ gm

$= 67.2$ L

8. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be

(A) 4 (B) 2 (C) 1 (D) 0.5 **[JEE 2005]**

Ans. (B)

Sol. $\frac{r_{He}}{r_{CH_4}} = \sqrt{\frac{16}{4}} = 2$

9. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is **[JEE 2009]**

Ans. (4)

Sol. $(u_{rms})_x = (u_{mp})_y$

$\frac{\sqrt{3RT_x}}{M_x} = \frac{\sqrt{2RT_x}}{M_Y}$

$\frac{3 \times 400}{40} = \frac{2 \times 60}{M_y}$

$M_y = \frac{2 \times 60 \times 40}{3 \times 400} = 4$

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Important Notes

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